# VECTO 4.x

Vehicle Energy Consumption Calculation Tool

## Release Notes

# VECTO 4.0.0.3210

## 16.10.2023

Vehicle Energy Consumption Calculation Tool

# Release Notes

# First official VECTO release for the 2nd Amendment of Regulation (EU) 2017/2400

TUG

# Features

- Declaration Mode
  - xEV-Heavy Lorries
  - Conventional Medium Lorries
  - xEV Medium Lorries
  - Conventional Buses
  - xEV Buses
- Dedicated user interface for simulating buses in Declaration Mode using the multistep approach: VECTOMultistep.exe
- Engineering Mode
  - xEV-Vehicles
- Updated XML-Reports and Vehicle Information Files to support new Vehicle Architectures
  - New Manufacturer Record File (MRF):
  - New Customer Information File (CIF):
  - Multistep Output (VIF):
- Updated simulation output (.vsum, .vmod)
- Updated Hashing Tool to handle new powertrain components (electric motor, battery, supercap ...)

- Supports .NET Framework 4.8 and .NET 6.0

# Features – Tool modes

| | Medium Lorries (XML) | Heavy Lorries (XML) | Primary buses (XML) | Interim buses (GUI) | Interim buses (XML) | Completed buses (JSON v7) | Completed buses (GUI) | Completed buses (XML) | Complete buses (JSON v10) | Complete buses (XML) |
|---|---|---|---|---|---|---|---|---|---|---|
| VECTO.exe | X | X | X | | | X | | X | X | X |
| VECTOcmd.exe | X | X | X | | | X | | X | X | X |
| VECTOMultistep.exe | X | X | X | X | X | X | X | X | X | X |
| Comments | | | | function: generates updated VIF | | | | | "Single step" manufacturing process | |

X … Simulation only
X … Simulation and editing

# Changes

- XML Job Files < v2.4 are no longer supported
- Dropped support for .NET Framework 4.5 (EOL 04/2022)

# Implementation of Declaration Mode for xEV-Vehicles

- Generic auxiliary parametrisation
- Generic parametrisation of usable SOC range
- Generic parametrisation of Serial/Parallel Hybrid Strategy parameters
- Automated simulation of OVC-HEV
  - Charge depleting mode (CD): The propulsion energy is provided by the electric storage only
  - Charge sustaining mode (CS): The propulsion energy is provided by the fuel storage

# Implementation of Declaration Mode for xEV-Lorries

- Generic E-PTO for PEV and S-HEV in Declaration Mode
- E-PTO for PEV and S-HEV in Engineering Mode

# Implementation of Declaration Mode for Buses

- Factor Method for conventional and xEV-Buses
- Updated bus auxiliary model